

MEMSYST STAGE 15 DEVELOPMENT MASTER PROMPT

Mobile Apps, Self-Service Portals & Omnichannel Experience Platform (MSSOEP)

You are a Senior Digital Experience Architect, Mobile Platform Architect, Membership Technology Architect, Enterprise SaaS Architect, and Principal Engineer responsible for building Stage 15 of the MemSyst ecosystem.

Stages 1–14 have already established:

Stage 1

Business Operations Platform

Stage 2

Identity & Access Platform

Stage 3

Tenant Management Platform

Stage 4

Membership & Apprentice Platform

Stage 5

Financial Infrastructure Platform

Stage 6

Communication & Engagement Platform

Stage 7

Digital Identity & Credential Platform

Stage 8

Marketplace, Business Directory & Commerce Platform

Stage 9

Training, Learning, Apprenticeship & Skills Development Platform

Stage 10

Support Hub, Opportunities, Grants, Loans & Member Services Platform

Stage 11

Events, Conferences, Meetings & Attendance Management Platform

Stage 12

Governance, Elections, Voting & Decision Management Platform

Stage 13

Analytics, Intelligence, Reporting & Decision Support Platform

Stage 14

Integration, API Gateway & Ecosystem Connectivity Platform

Stage 15 introduces the unified digital experience layer of MemSys.

This stage brings every platform capability directly to members, apprentices, executives, administrators, training centers, branches, and regions through modern web, mobile, and self-service experiences.

STAGE 15 OBJECTIVE

Build the:

Mobile Apps, Self-Service Portals & Omnichannel Experience Platform (MSSOEP)

Purpose:

Provide:

Mobile Apps

Self-Service Portals

Progressive Web Apps

Executive Portals

Branch Portals

Regional Portals

Member Portals

Apprentice Portals

Unified User Experience

across the entire MemSys ecosystem.

CRITICAL EXPERIENCE PRINCIPLES

Principle 1

Mobile First

All major platform functions must be accessible from mobile devices.

Members should not need desktop access for common tasks.

Principle 2

Role-Based Experiences

Users only see functionality relevant to their role.

Examples:

Member
Apprentice
Executive
Branch Officer
Regional Officer
Training Center
Tenant Administrator
Platform Administrator

Principle 3

Tenant Branding

Each tenant may customize:

Logo
Colors
Homepage
Menus
Terminology
Public Content

while operating on the shared MemSyst infrastructure.

Principle 4

Consistent Experience

Users should experience one platform regardless of module.

Navigation, notifications, identity, and workflows must remain consistent.

STAGE 15 DELIVERABLES

Build:

Mobile Application Platform

Progressive Web App Platform

Member Self-Service Portal

Executive Portal

Branch Portal

Regional Portal

Tenant Portal

Experience Personalization Engine

MODULE 1

MOBILE APPLICATION PLATFORM

Purpose:

Provide native-like mobile experiences.

Platforms

Android

iOS

Progressive Web App

Features

Login

Member Dashboard

Notifications

Wallet Access

Marketplace Access

Events

Training

Support Hub

Digital ID Cards

MODULE 2

MEMBER SELF-SERVICE PORTAL

Purpose:

Enable members to manage their accounts independently.

Features

Profile Management
Membership Status
Renewals
Payments
Wallet
Marketplace
Training
Events
Support Applications
Certificates
Digital ID Cards

MODULE 3

APPRENTICE PORTAL

Purpose:

Provide apprentice-specific functionality.

Features

Profile
Training Progress
Mentor Information
Assessments
Attendance
Certificates
Upgrade Eligibility
Notifications

Restrictions

Apprentices can only access their own records.

MODULE 4

EXECUTIVE PORTAL

Purpose:

Provide leadership-focused functionality.

Features

Approvals

Governance Activities

Membership Reviews

Reports

Analytics

Events

Committee Activities

Communications

MODULE 5

BRANCH PORTAL

Purpose:

Provide branch-level operational management.

Features

Branch Dashboard

Member Management

Approvals

Branch Wallet

Events

Reports

Communications

Marketplace Oversight

MODULE 6

REGIONAL PORTAL

Purpose:

Provide regional oversight.

Features

Regional Dashboard

Branch Monitoring

Approvals

Regional Wallet

Reports

Events

Training Oversight

MODULE 7

TRAINING CENTER PORTAL

Purpose:

Support accredited training centers.

Features

Student Management

Course Management

Attendance

Assessments

Graduation Recommendations

Reports

MODULE 8

DIGITAL WALLET EXPERIENCE

Purpose:

Provide financial self-service.

Features

Balance
Transactions
Payments
Receipts
Transfers
Billing History

Integration

Must connect with:

Stage 5 Financial Infrastructure

MODULE 9

DIGITAL ID EXPERIENCE

Purpose:

Provide portable identity.

Features

Digital ID Card
QR Verification
Credential Wallet
Certificates
Executive Credentials

Integration

Must connect with:

Stage 7 Credential Platform

MODULE 10

PERSONALIZED DASHBOARD ENGINE

Purpose:

Provide customized user experiences.

Dashboard Content

Based on:

Role

Permissions

Tenant

Region

Branch

Activity History

Examples

Members see:

Membership

Wallet

Events

Marketplace

Training

Executives see:

Approvals

Analytics

Governance

Reports

MODULE 11

OMNICHANNEL NOTIFICATION CENTER

Purpose:

Provide unified communication visibility.

Channels

Email

SMS

Push
In-App

Features

Notification History
Read Status
Preferences
Subscriptions

MODULE 12

SEARCH & DISCOVERY ENGINE

Purpose:

Provide unified search.

Search Targets

Members
Businesses
Events
Courses
Opportunities
Certificates
Documents

Requirements

Search results must respect permissions.

MODULE 13

OFFLINE EXPERIENCE FOUNDATION

Purpose:

Prepare mobile users with limited connectivity.

Features

Offline Viewing
Local Caching
Background Sync
Queued Actions

Priority

Digital IDs and credentials should remain accessible offline.

MODULE 14

EXPERIENCE PERSONALIZATION ENGINE

Purpose:

Allow tenant-specific customization.

Customizable Elements

Branding
Menus
Landing Pages
Terminology
Widgets
Themes

Control

Managed by MemSyst and tenant administrators.

MODULE 15

MOBILE EVENT EXPERIENCE

Purpose:

Enhance Stage 11 event participation.

Features

Event Registration
QR Tickets
Check-In
Schedules
Certificates

MODULE 16

MOBILE GOVERNANCE EXPERIENCE

Purpose:

Enhance Stage 12 participation.

Features

Nominations
Voting
Election Results
Governance Notifications
Committee Activities

MODULE 17

MOBILE MARKETPLACE EXPERIENCE

Purpose:

Enhance Stage 8 participation.

Features

Business Profiles
Marketplace Listings
Opportunity Applications
Directory Search

MODULE 18

MOBILE TRAINING EXPERIENCE

Purpose:

Enhance Stage 9 participation.

Features

Course Enrollment

Attendance

Assessments

Certificates

Progress Tracking

MODULE 19

USER PREFERENCE MANAGEMENT

Purpose:

Allow users to control their experience.

Settings

Notifications

Privacy

Language

Theme

Communication Preferences

MODULE 20

EXPERIENCE ANALYTICS ENGINE

Purpose:

Measure platform adoption.

Metrics

Active Users

Session Duration

Feature Usage
Mobile Usage
Portal Usage
Engagement Trends

MODULE 21

EXPERIENCE AUDIT ENGINE

Purpose:

Track user interactions.

Audit Events

Login
Profile Update
Payment Initiated
Vote Cast
Event Registration
Certificate Viewed

DATABASE DESIGN REQUIREMENTS

Before implementation:

Design complete entities for:

User Preferences
Device Registrations
Mobile Sessions
Notification Preferences
Dashboard Configurations
Portal Configurations
Experience Analytics

Define:

Relationships
Indexes
Caching Models
Offline Sync Rules
Security Rules

SECURITY REQUIREMENTS

Integrate fully with:

Stage 2 Identity Platform

Stage 7 Credential Platform

Enforce:

RBAC

Tenant Isolation

Device Security

Session Management

Audit Logging

PERFORMANCE REQUIREMENTS

Support:

Thousands of Concurrent Users

Mobile Networks

Low-Bandwidth Environments

Offline Synchronization

Push Notifications

FUTURE READINESS REQUIREMENTS

Stage 15 must prepare for:

AI Assistants

Voice Interfaces

Chat-Based Operations

Biometric Authentication

Super Apps

Wearables

Digital Government Integration

Do not implement these now.

Design all supporting foundations.

ARCHITECTURE REQUIREMENTS

Use:

Mobile-First Architecture
API-Driven Experience Layer
Offline-First Principles
Role-Based Experience Design
Tenant-Aware Branding

Avoid:

Hardcoded User Experiences
Role Leakage
Tenant Branding Restrictions
Desktop-Only Features

SUCCESS CRITERIA

Stage 15 is complete only when:

- ✓ Mobile applications are operational
- ✓ Progressive Web Apps are operational
- ✓ Member self-service portals exist
- ✓ Apprentice portals exist
- ✓ Executive portals exist
- ✓ Branch portals exist
- ✓ Regional portals exist
- ✓ Training center portals exist
- ✓ Digital wallet experience exists
- ✓ Digital ID experience exists
- ✓ Personalized dashboards exist
- ✓ Omnichannel notification center exists
- ✓ Unified search exists
- ✓ Tenant branding exists
- ✓ Experience analytics exist

✓ Experience audit history exists

✓ Platform is fully prepared for the final stage: **Stage 16 – MemSyst Ecosystem, Automation, AI & Future Innovation Platform**

Build this stage as the primary user experience layer of the MemSyst ecosystem, ensuring every stakeholder can access platform services seamlessly from any device, anywhere, at any time.